

A zwitterionic terpolymer based coating potentially for antibacterial and antifouling applications

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HIGHLIGHTS

- The novel P(VIm-co-HPMA-co-BA) were synthesized via free radical polymerization for the first time.
- P(VPS-co-HPMA-co-BA) were prepared via “one-pot” zwitterionization.
- P(VPS-co-HPMA-co-BA) membranes had excellent anti-bacterial and anti-adhesion properties.
- P(VPS-co-HPMA-co-BA) membranes exhibited non-toxicity property.

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ABSTRACT

Antifouling materials are highly demanded in global marine market due to increasingly aggravated marine biofouling problem and unbalanced marine ecology. Herein, poly (vinyl imidazole-co-hydroxypropyl methacrylate-co-butyl acrylate) (P(VIm-co-HPMA-co-BA)) was prepared via free radical polymerization of commercially available monomers of vinyl imidazole (VIm), hydroxypropyl methacrylate (HPMA) and butyl acrylate (BA). Then the zwitterionic terpolymer (P(VPS-co-HPMA-co-BA)) was obtained by ionicization of P(VIm-co-HPMA-co-BA) with 1, 3-propionate lactone (1, 3-PS) in EtOH solutions. Furthermore, the membrane samples, P(VIm-co-HPMA-co-BA) and P(VPS-co-HPMA-co-BA) were prepared by tape casting method with DI water bath. Significantly, P(VPS-co-HPMA-co-BA) membranes have shown better bactericidal activity against Gram-negative *Escherichia coli* (*E. coli*) and Gram-positive *Staphylococcus aureus* (*S. aureus*) with a mean area size of 3.5 cm². Meanwhile, the bacterial adhesion assay showed that P(VPS-co-HPMA-co-BA) membranes were found the absence of adherent bacteria, and exhibited excellent anti-adhesion property with the increase of molar ratios of PVPS by the observation of bacteria population density. The mechanical testing results indicated that the molar ratios of 65/10/25 in P(VIm-co-HPMA-co-BA) membranes and P(VPS-co-HPMA-co-BA) membranes had high breaking elongation ratio and great tensile strength. Specifically, the maximum tensile strength of P(VPS-co-HPMA-co-BA) membrane could reach up to 10.9 MPa. The results showed that the novel P(VPS-co-HPMA-co-BA) membrane exhibited both anti-bacterial, anti-adhesion property and mechanical performance, manifesting its prospectively potential applications in new highly efficient and eco-friendly marine antifouling coatings.

1. Introduction

The great majority of biological fouling have been caused by the bioaccumulation of dominant components of organic components and microbes [1–3]. The biofilms, generated by microorganisms adhesion in ocean, have caused the overuse of detergents, low mechanical property of coatings and the additional cost of device maintenances [4–7]. In

order to prevent ocean devices from the adhesion of micro-organisms, excellent properties of marine coatings that possess anti-adhesion, anti-bacterial property, high mechanical strength and environmentally friendly performance, come into being and are in pressing requirements in the marine market [8]. Therefore, it is of great application and research value to develop materials with anti-bacterial and anti-adhesion properties, especially antifouling materials, degradable

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materials (polyester-polyurethane (PET-PU), modified polyester and modified polyacrylate with excellent antifouling activity and comprehensive mechanical performance, N-(2,4,6-trichlorophenyl)-maleimide that was grafted to polydimethylsiloxane-polyurethane (PDMS-PU)) with a strong inhibitory effect on bacteria, diatoms, etc. [9], polyethylenimine-methyl sulfonate betaine (PET-SBMA) bearing amide groups with the function of preparing antifouling membranes surface materials, prevent living bacteria from adhering and clean up the dead or the injured bacteria for specific application in marine areas, it is vitally significant to stress that zwitterionic polymers are promising antifouling materials based on previous study [10–13].

Poly vinyl imidazole (PVIIm) exhibited excellent antibacterial property and had widely applications in polysaccharides by chemical grafting polymerization, such as chitosan [14] and carboxymethyl chitosan (CMCh) [15], PVIIm grafted onto chitosan (Chitosan-g-PVIIm), the graft copolymerization of vinyl imidazole (VIm) onto CMCh (CMCh-g-PVIIm), enhanced water solubility of chitosan and CMCh, and endowed them with biocompatibility and biodegradable property. According to the previous reports, PVIIm derivatives possessed excellent hydrophilic property and anti-bacterial activity, but were short-term of stability and long-lasting anti-bacterial activity. Until recently, the reason for the excellent anti-adhesion property of zwitterionic polymer that has attracted great attention is that it forms a layer of hydration by electrostatic interaction, hydrogen bonding between equimolar positive and negative charges and water molecules [16]. The grafted zwitterionic polymers, such as poly(carboxybetainemethacrylate) (PCBMA)-grafted surfaces (PCBMA-g-glass), had better chemical stability, good hydrophilicity and the activity of reducing adhesion of microorganisms on the material surfaces [17], however, it had no anti-bacterial property [10, 18–22]. Radical polymerization is the widest industrial method of the preparation of kinds of polymers, atom transfer radical polymerization (ATRP) has become an appropriate choice by the virtue of its “live”/controlled radical polymerization, however, some drawbacks of ATRP include the cumbersome extraction of transition metal complexes used as catalysts and complicated preparation process [23–25], and the distinct advantages of free radical polymerization based on two or more monomers include the “one-pot” preparation that grant fabrication process easy and feasible activities, enhanced molecular weight and simplicity of purifying [26,27]. Recently, poly (sulfobetaine methacrylate-*b*-[2-(Methacryloyloxy) ethyl] trimethylammonium chloride solution) (P(SBMA-*b*-MTAC)) were grafted on the TFC membranes surface via activators prepared by surface-initiated ATRP techniques, which showed great bactericidal ability and anti-adhesion property [18], but the preparation methods were complicated and hard to industrialize. Based on the above viewpoints, it is necessary to synthesis a kind of marine coatings with dual anti-bacterial and anti-adhesion properties, furthermore, the novel materials should possess the non-toxicity property and the function of facile preparation method.

VIm easily forms PVIIm of high molecular weights for resolving the drawback of PVIIm derivatives (Chitosan-g-PVIIm, CMCh-g-PVIIm) mentioned above, the zwitterionized PVIIm (PVPS) could enhance antimicrobial ability [22,24]. And BA, its homopolymer poly(butyl acrylate) (PBA) bearing hydrocarbon chain, was applied to be a ductile and flexible polymer network [28,29] and exhibited effective and stable bactericidal performance [30,31] as well as non-toxicity property in vitro experiments with fibroblast-like cells (L929) [28] and prime endothelial progenitor cells [32] to reduce the cytotoxicity of zwitterionic polymers. However, there are some challenges about the incorporation and reaction efficiency of two or more monomers in the free radical polymerization process that need to be solved, poly (2-hydroxypropyl methacrylate) (PHPMA) had great hydrophobic nature [33,34] and forming ability, PHPMA/SiO₂ nanocomposite particles had great formation efficiency owing to the high-speed polymerization of 2-hydroxypropyl methacrylate (HPMA) at 60 °C [35]. Hence, choosing above three monomers (VIm, HPMA and BA), “one-pot” synthesis of terpolymers (P(VIm-co-HPMA-co-BA)) were gained via free

radical polymerization for integrating with their merits, and zwitterionic polymers (P(VPS-co-HPMA-co-BA)) were prepared by 1, 3-PS. P (VIm-co-HPMA-co-BA) membranes were prepared via tape casting method and P(VPS-co-HPMA-co-BA) membranes were zwitterionized by 1, 3-PS EtOH solutions [36]. The above samples were characterized by ¹H NMR, FT-IR, GPC and Element Analyzer, the whole preparation process possessed many merits, such as facial, low-cost and environment-friendly preparation. The anti-bacterial activity and anti-adhesion property of P(VPS-co-HPMA-co-BA) membranes were evaluated by the antibacterial test and bacterial adhesion assay, the surface hydrophilicity property, quaternization degree and quaternization density properties as well as mechanical performance were illustrated in details. Furthermore, the cytotoxicity of P (VIm-co-HPMA-co-BA) and P(VPS-co-HPMA-co-BA) membranes were analyzed against NIH-3T3 cells by the cytotoxicity experiments, which had a potential application in marine paints and coatings.

2. Experimental

2.1. Materials

Vinyl imidazole (VIm, 99%), butyl acrylate (BA, 50 ppm MEHQ stabilizer), hydroxypropyl methacrylate (HPMA, 50 ppm MEHQ stabilizer), 1, 3-propionic lactone (1, 3-PS, 99%); Ether (analytical grade) were purchased from Aladdin Biochemical Technology Company (Shanghai, China); Particularly, the MEHQ stabilizer was removed by chromatographed on silica with dichloromethane/ethyl acetate (2:1) and restored in an argon-filled environment. 2, 2'-azobis (2-methylpropionitrile) (AIBN, 0.2 M in toluene); N, N-Dimethyl amide (DMF, analytical grade); Dimethyl sulfoxide (DMSO, analytical grade), anhydrous magnesium sulfate (MgSO₄, AR grade), EtOH (analytical grade) were obtained from Sinopharm Reagent Company (Shanghai, China); PBS (KCl 0.5 g, KH₂PO₄ 0.5 g, NaCl 20 g, Na₂HPO₄ 2.8647 g, 500 mL), O-Nitrophenyl-β-D-galactopyranoside (ONPG, 99%) were obtained from McLean biochemical technology company (Shanghai, China), Dulbecco's modified Eagle's medium and Cell Counting Kit-8 were acquired from Gibco Technology (Carlsbad, CA, USA) and Dojindo Molecular Technology company (Shanghai, China), respectively. All other reagents were analytical grade and were used without refining ulteriorly.

2.2. Synthesis of VPS and P(VIm-co-HPMA-co-BA)

The synthetic root of VPS was reported in previous literature [37]. ¹H NMR (δ = 5.40, 5.79 (d, e, -H₂C=C), δ = 7.10 (f, HC-N near imidazole ring), δ = 9.00 (i, N-CH-N⁺), δ = 7.54 (g, CH-N), δ = 7.70 (h, CH-N⁺), δ = 2.26 (a, C-CH₂-C), δ = 2.87 (b, CH₂-S), δ = 4.32 (c, CH₂-N⁺), δ = 4.70 (d₂O) [37].

The P(VIm-co-HPMA-co-BA) were synthesized by free radical polymerization of desired amount of monomers. Detailly, 6.79 mL (0.075 mol) VIm, 1.70 mL (0.012 mol) HPMA and 2.16 mL (0.015 mol) BA, which were added into 16 mL EtOH solution using 0.0132 g (0.08 mmol) AIBN as initiator, were carried out for 24 h at 65 °C under anhydrous and oxygen-free atmosphere. Especially different compositions of P(VIm-co-HPMA-co-BA) were synthesized via free radical polymerization with the molar ratios of VIm/HPMA/BA = 50/20/20; 60/20/30; 65/10/25; 75/10/15. P(VIm-co-HPMA-co-BA) was purified by ether and dialyzed in DI water for 2 days, and then dried via freeze-drying. ¹H NMR: δ = 0.87 (a, -CH₃ group), δ = 1.23–1.29 (b, c, h, j, -CH₂ and -CH group), δ = 6.85–7.51 (e, f, g, CH in VIm), δ = 4.95 (d, -CH₂), δ = 3.79 (i, -OH), δ = 2.50 (DMSO-*d*₆), the structure of each monomer were detailly showed in Fig.2

2.3. Synthesis of P(VPS-co-HPMA-co-BA)

0.8 g P(VIm-co-HPMA-co-BA) were dissolved in 10 mL DMF under argon atmosphere, and then P(VPS-co-HPMA-co-BA) were

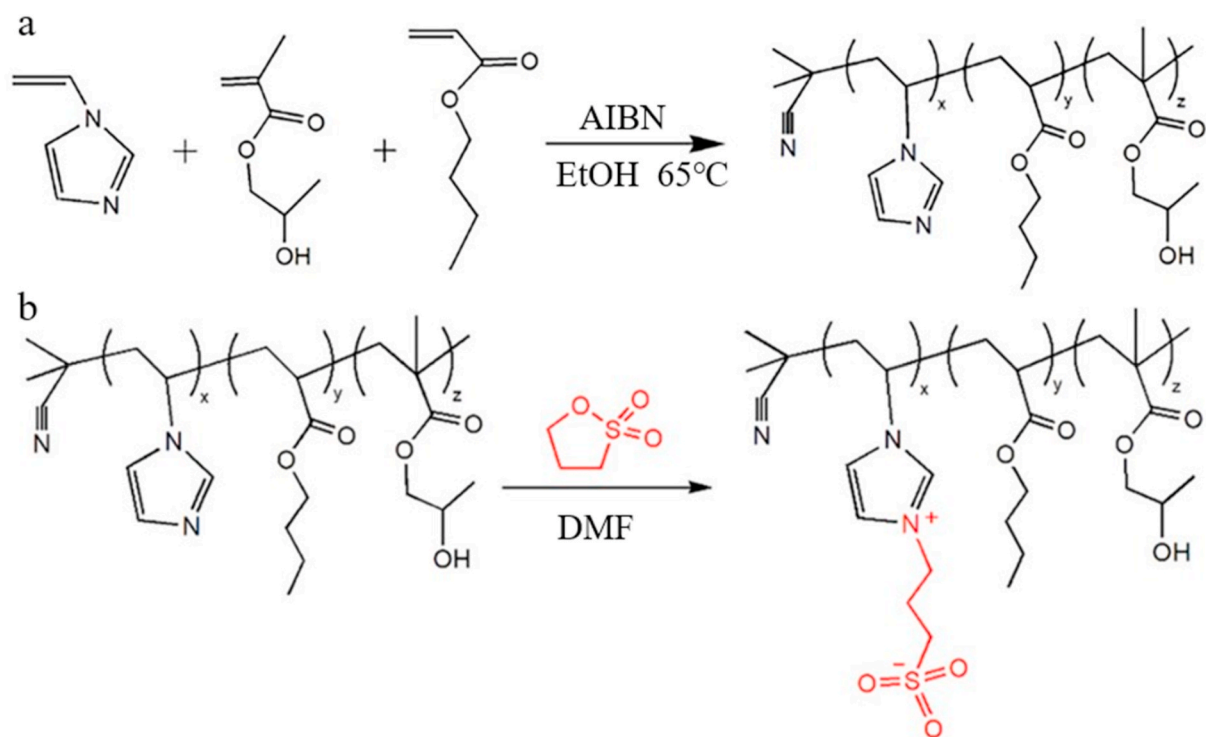


Fig. 1. The synthesis scheme of P(Vim-co-HPMA-co-BA) and P(VPS-co-HPMA-co-BA).

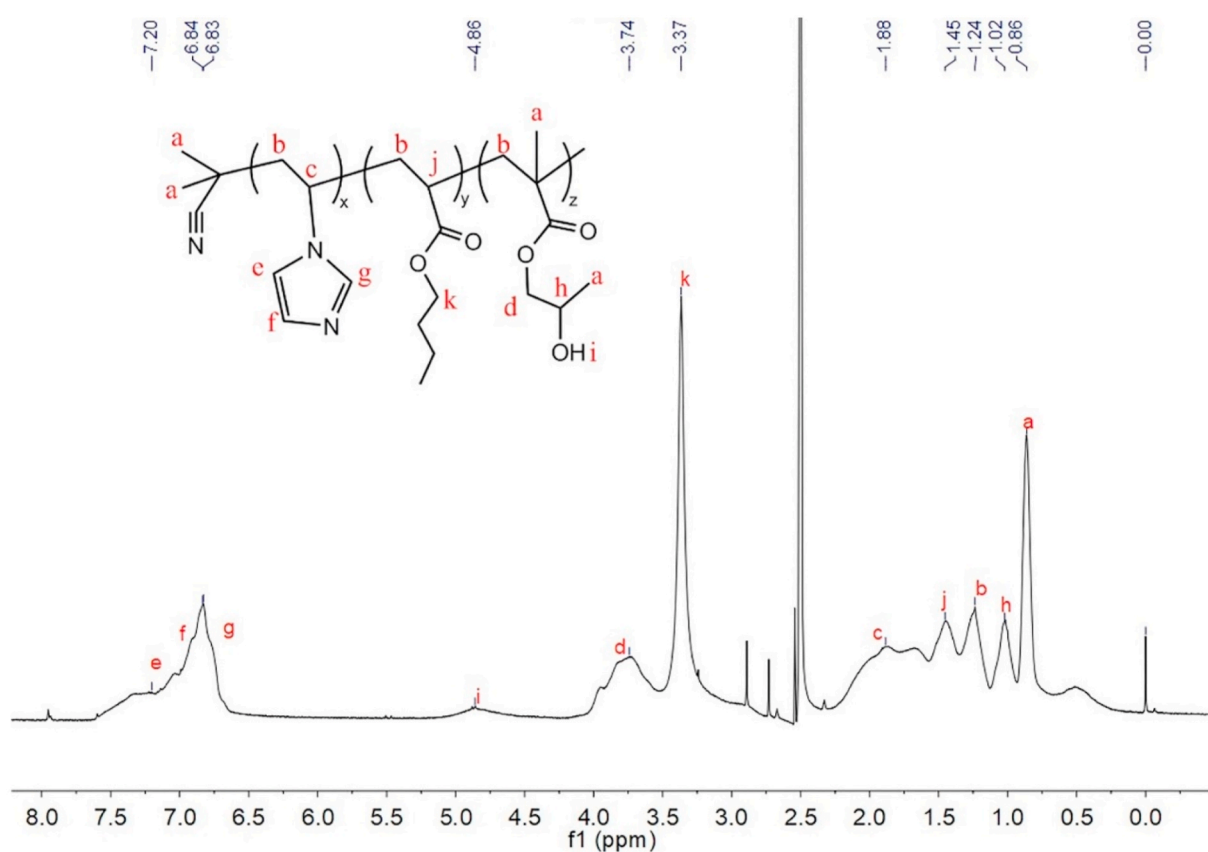


Fig. 2. The ¹H NMR spectra of P(Vim-co-HPMA-co-BA).

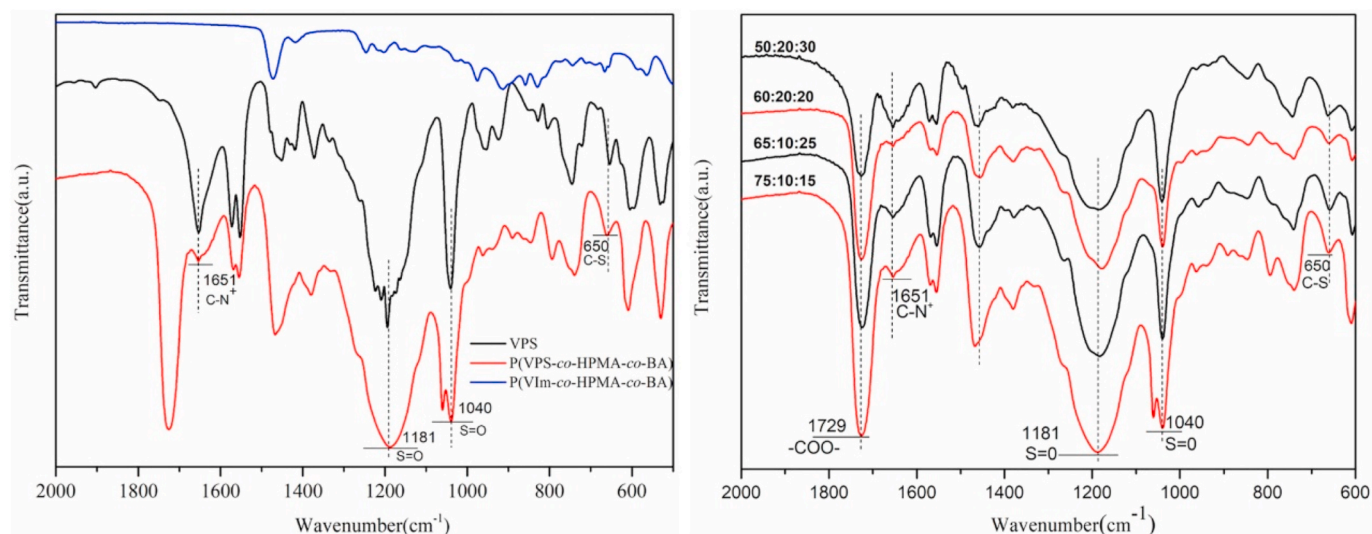


Fig. 3. The FT-IR spectra of VPS, P(VIm-co-HPMA-co-BA), P(VPS-co-HPMA-co-BA) (a) and P(VPS-co-HPMA-co-BA) with different molar ratios of VPS/HPMA/BA (b).

Table 1

The name of P(VIm-co-HPMA-co-BA) with different molar ratios of VIm/HPMA/BA.

Polymer	Feed		Sample
	VIm/HPMA/BA	AIBN/[VIm]+[HPMA]+[BA]	
P(VIm ₅₀ -co-HPMA ₂₀ -co-BA ₃₀)	50/20/30	1/1280	P(VIm ₅₀ -co-HPMA ₂₀ -co-BA ₃₀)-I
P(VIm ₆₀ -co-HPMA ₂₀ -co-BA ₂₀)	60/20/20	1/1280	P(VIm ₆₀ -co-HPMA ₂₀ -co-BA ₂₀)-II
P(VIm ₆₅ -co-HPMA ₁₀ -co-BA ₂₅)	65/10/25	1/1280	P(VIm ₆₅ -co-HPMA ₁₀ -co-BA ₂₅)-III
P(VIm ₇₅ -co-HPMA ₁₀ -co-BA ₁₅)	75/10/15	1/1280	P(VIm ₇₅ -co-HPMA ₁₀ -co-BA ₁₅)-IV
P(VIm ₅₀ -co-HPMA ₂₀ -co-BA ₃₀)	50/20/30	1/1920	P(VIm ₅₀ -co-HPMA ₂₀ -co-BA ₃₀)-V
P(VIm ₆₀ -co-HPMA ₂₀ -co-BA ₂₀)	60/20/20	1/1920	P(VIm ₆₀ -co-HPMA ₂₀ -co-BA ₂₀)-VI
P(VIm ₆₅ -co-HPMA ₁₀ -co-BA ₂₅)	65/10/25	1/1920	P(VIm ₆₅ -co-HPMA ₁₀ -co-BA ₂₅)-VII
P(VIm ₇₅ -co-HPMA ₁₀ -co-BA ₁₅)	75/10/15	1/1920	P(VIm ₇₅ -co-HPMA ₁₀ -co-BA ₁₅)-VIII

Table 2

The molecular weights, PDI and elemental analysis of samples.

Sample	Mn (g/mol)	PDI	Yield (%)	composition				The proportion of P(VIm) in terpolymer	Terpolymer formula
				%C	%H	%N			
P(VIm ₅₀ -co-HPMA ₂₀ -co-BA ₃₀)-I	193300	2.04	90.8	60.5	7.88	10.50	44.88%		C _{6.72} H _{10.5} N _{1.0}
P(VIm ₆₀ -co-HPMA ₂₀ -co-BA ₂₀)-II	217000	1.80	93.4	59.6	7.86	11.39	48.33%		C _{6.10} H _{9.66} N _{1.0}
P(VIm ₆₅ -co-HPMA ₁₀ -co-BA ₂₅)-III	193800	2.16	92.4	61.8	7.95	10.49	44.31%		C _{6.87} H _{10.61} N _{1.0}
P(VIm ₇₅ -co-HPMA ₁₀ -co-BA ₁₅)-IV	151000	1.61	92.5	61.2	7.67	15.10	60.20%		C _{4.73} H _{7.113} N _{1.0}
P(VIm ₅₀ -co-HPMA ₂₀ -co-BA ₃₀)-V	219100	2.08	93.8	60.8	7.28	8.15	37.22%		C _{8.74} H _{12.55} N _{1.0}
P(VIm ₆₀ -co-HPMA ₂₀ -co-BA ₂₀)-VI	244400	2.07	95.2	60.5	7.70	11.98	50.23%		C _{5.89} H _{9.0} N _{1.0}
P(VIm ₆₅ -co-HPMA ₁₀ -co-BA ₂₅)-VII	144800	1.37	95.6	62.3	7.78	11.93	49.34%		C _{6.10} H _{9.15} N _{1.0}
P(VIm ₇₅ -co-HPMA ₁₀ -co-BA ₁₅)-VIII	151900	1.64	91.2	63.3	8.05	15.89	61.34%		C _{4.65} H _{7.09} N _{1.0}

zwitterionized by 1, 3-PS for 6 h at 70 °C, finally, P(VPS-co-HPMA-co-BA) were purified by acetone and dried in vacuum oven 24 h.

2.4. Preparation of P(VIm-co-HPMA-co-BA) and P(VPS-co-HPMA-co-BA) membranes

The P(VIm-co-HPMA-co-BA) membranes were prepared via tape casting method. Detailedly, 2 g P(VIm-co-HPMA-co-BA) were dissolved in 5 mL DMF, and then were added dropwisely to the glass slide, which were immersed in coagulation bath using deionized water to remove DMF to obtain P(VIm-co-HPMA-co-BA) membranes. Further, the modified membranes were formed via zwitterionization by dipping 1, 3-PS EtOH solutions into P(VIm-co-HPMA-co-BA) membranes surface for 48 h at 50 °C. Finally, the modified membranes were cleaned with deionized water.

2.5. Characterization

¹H NMR of VPS and P(VIm-co-HPMA-co-BA) were measured on Bruker DMX-400 NMR spectrometer. Molecular weight and polydispersity index (PDI) of P(VIm-co-HPMA-co-BA) were measured using gel permeation chromatography (GPC, Waters 1515, American waters). The elements (C, H, N) of P(VIm-co-HPMA-co-BA) were determined on Element Analyzer (VARIO ELIII, ELEMENTAR, Germany). The FT-IR spectrum of VPS, P(VIm-co-HPMA-co-BA) and P(VPS-co-HPMA-co-BA) were analyzed on Nicolet Magna-IR 550 (Thermo Electron, American), the scanning range was 4000-400 cm⁻¹ and scanning rate was 65 times s⁻¹.

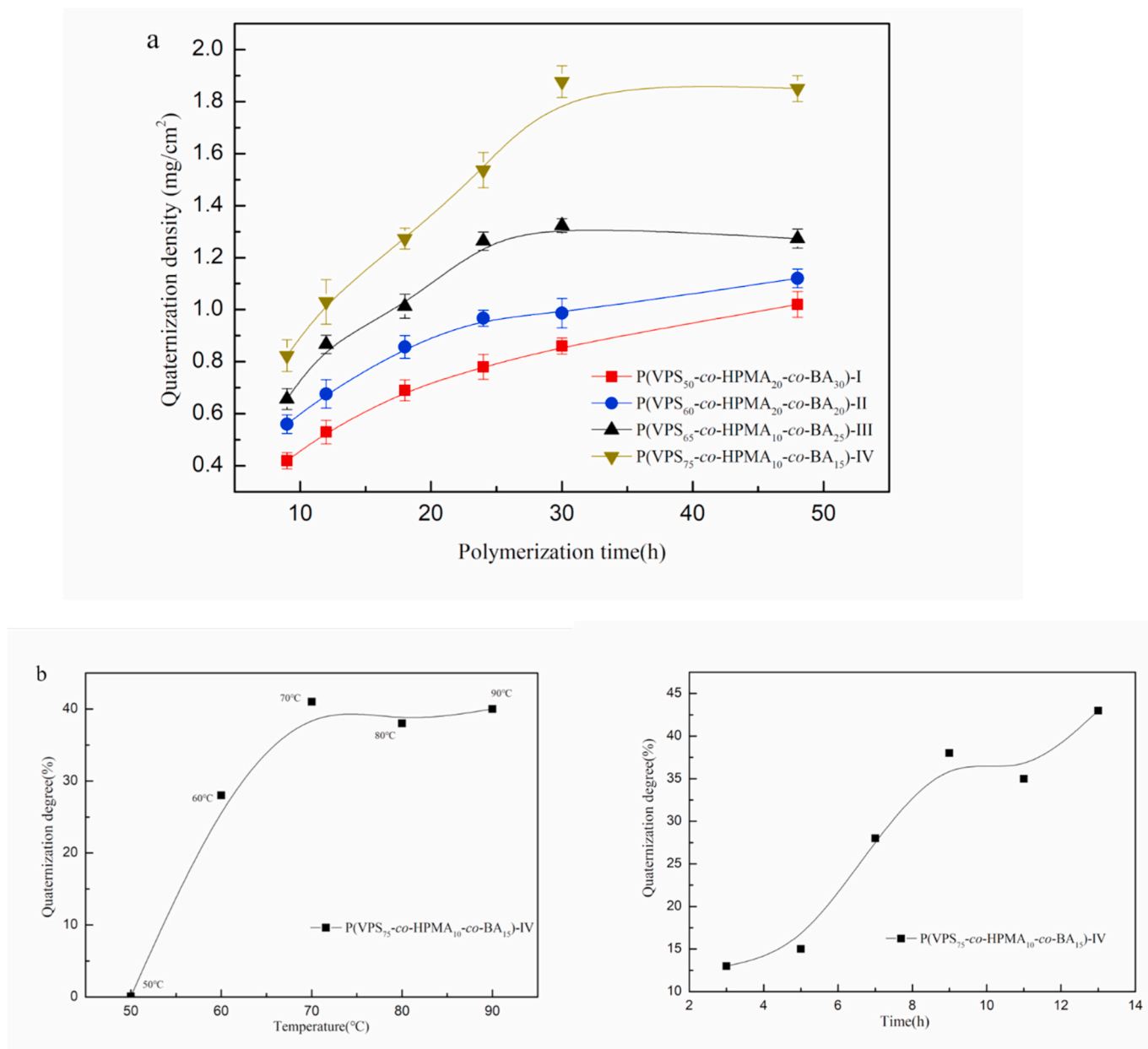


Fig. 4. The QE (a) and QD (b) of samples were to analyze process of zwitterionicalization.

2.5.1. Contact angle measurement

The surface hydrophilicity of P(VIm-co-HPMA-co-BA) and P(VPS-co-HPMA-co-BA) membranes placed in glass slide were estimated by contact angle measuring instrument (JC2000D2, Powereach, China). The operations were carried out on three spots of membranes samples to obtain average value.

2.5.2. Quaternization degree and quaternization density

The zwitterionicalized P(VIm-co-HPMA-co-BA) membranes were mainly characterized by quaternization degree (QD) and quaternization density (QE) [38], which were measured by weighing method. The first way to represent QD of P(VPS-co-HPMA-co-BA) was calculated by formula (1) [39]. The second way to illustrate QE of P(VPS-co-HPMA-co-BA) membranes was determined by the following formula (2) [26,36,40].

$$QD = \frac{M_2 - M_0}{M_0 \times 100} \quad (1)$$

$$QE = \frac{m_1 - m_0}{A} \quad (2)$$

Where M_0 (mg), M_2 (mg), m_0 (mg), m_1 (mg) were the weight of P(VIm-co-HPMA-co-BA), P(VPS-co-HPMA-co-BA), P(VIm-co-HPMA-co-BA) and P(VPS-co-HPMA-co-BA) membranes, respectively, which were rinsed completely with deionized water and dried overnight until the constant-weight membranes were gained after repeat operations, A (cm²) represented the area of membranes and the area of membranes were 3.5 cm², all samples were operated twice to obtain the mean values.

2.5.3. Mechanical properties

The tensile strength and breaking elongation ratio of P(VIm-co-HPMA-co-BA), P(VPS-co-HPMA-co-BA) membranes were tested on universal tension machine (2K WCMT 2203, New sansi material Co., Ltd., China) at a drawing rate of 2 mm min⁻¹ at a room temperature, the membranes were cut into standard dumbbell-shaped splines with a gauge length of 25 mm. Samples were performed to obtain an average

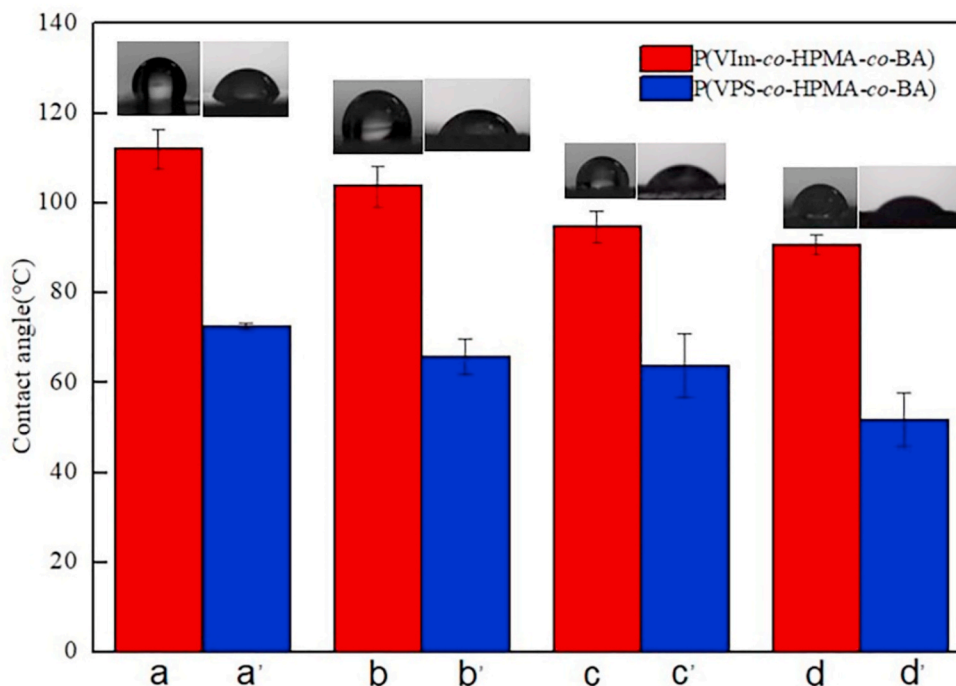


Fig. 5. The water contact angle of P(VIm-co-HPMA-co-BA) and P(VPS-co-HPMA-co-BA) membranes (P(VIm₅₀-co-HPMA₂₀-co-PBA₃₀)-I (a); P(VPS₅₀-co-HPMA₂₀-co-BA₃₀)-I (a'); P(VIm₆₀-co-HPMA₂₀-co-BA₂₀)-II (b); P(VPS₆₀-co-HPMA₂₀-co-BA₂₀)-II (b'); P(VIm₆₅-co-HPMA₁₀-co-BA₂₅)-III (c); P(VPS₆₅-co-HPMA₁₀-co-BA₂₅)-III (c'); P(VIm₇₅-co-HPMA₁₀-co-BA₁₅)-IV (d); P(VPS₇₅-co-HPMA₁₀-co-BA₁₅)-IV (d')).

value. The size of dumbbell spline was $50 \times 4 \text{ mm}^2$ (length $\times$ width), and broken elongation was measured as follow (3) [41].

$$\text{Elongation}(\%) = \frac{L_1 - L_0}{L_0} \times 100\% \quad (3)$$

where L_0 is the original length of the samples, and L_1 is the deformation length in the direction of the force.

2.6. Antibacterial properties assessments

The inhibition zone could qualitatively characterize antibacterial effect of VPS DI water solution (20 mg/mL) and P(VPS-co-HPMA-co-BA) membranes. *E. coli* and *S. aureus* were chosen to be representative bacterium during the whole test process. Detailly, 20 μL bacterial suspensions ($2 \times 10^5 \text{ CFU mL}^{-1}$) were transferred into a sterile culture dish containing solid medium, a filter paper loaded with VPS DI water solution (20 mg/mL) and P(VPS-co-HPMA-co-BA) membranes (3.5 cm^2) were then placed in the center of the medium in Petri dish (10 cm in diameter), which were cultivated in a constant-temperature incubator (37°C) for 24 h.

In the meantime, the antibacterial activities of P(VPS-co-HPMA-co-BA) membranes were quantitatively measured by Colony-counting Methods (CCM) [42,43]. In detail, bacterium strains were cultivated at 37°C overnight and diluted to a concentration of $2 \times 10^5 \text{ CFU mL}^{-1}$. The P(VPS-co-HPMA-co-BA) membranes (3.5 cm^2) and VPS DI water solution (20 mg/mL) were placed in five 15-mL conical flask consisted of 1 mL bacterial suspension and 9 mL PBS, separately, the flasks were then cultivated for 5 h at 37°C . The numbers of living bacterium were enumerated visibly and the sterilizing rate ($R(\%)$) were counted on the basis of formula (6) [44].

$$R(\%) = \frac{N_s - N_m}{N_s} \times 100\% \quad (6)$$

Where N_s , N_m represent the numbers of living bacterium from control samples and test samples, respectively. Additionally, the CCM was

operated on the super-clean bench, which was used for detecting the sterilization rates of samples.

2.7. Membrane permeability changes

The activity of P(VPS-co-HPMA-co-BA) membranes to permeabilize bacterium was measured by the ELIASA (ELX800 Bio Tek) (410-420 nm), owing to *o*-nitrophenol was produced because of 2-Nitrophenyl β -D-galactopyranoside (ONPG) stimulated by the beta-galactosidase that existed in bacteria. Detailly, 1.0 mL bacterial suspension ($2 \times 10^5 \text{ CFU mL}^{-1}$) was added into three test tubes containing 0.04 g 2-Nitrophenyl β -D-galactopyranoside and 10 mL PBS, respectively, two of which were cultivated with P(VPS-co-HPMA-co-BA) membranes (3.5 cm^2) in a constant-temperature incubator (37°C). The release of beta-galactosidase was characterized by the relationship between the OD₄₂₀ of culture solutions and incubation times.

2.8. Scanning electron microscopy

The morphology of bacteria was characterized by SEM. Briefly, the P(VPS-co-HPMA-co-BA) membranes were cultivated with *E. coli*, *S. aureus* bacterial suspensions ($2 \times 10^5 \text{ CFU mL}^{-1}$) and 10 mL PBS at 37°C for 4 h. After 4 h, the culture solutions were washed three times with PBS and fixed with glutaraldehyde PBS solution (4%, v/v) overnight at 25°C , the cultures were then rinsed with PBS and dehydrated with graded ethanol aqueous solution (30%, 50%, 70%, 90%, 95%, 100%), afterward, the samples were freeze-dried for 24 h. The samples were prepared and observed by Field-Emission Scanning Electron Microscope (Gemini SEM 500, Germany).

2.9. The bacterial adhesion assay

A 1 mL *E. coli* suspension ($2 \times 10^5 \text{ CFU mL}^{-1}$) was dripped into polystyrene Petri dish surface and P(VPS-co-HPMA-co-BA) membranes surface, which were cleaned with sterile water and covered with Petri

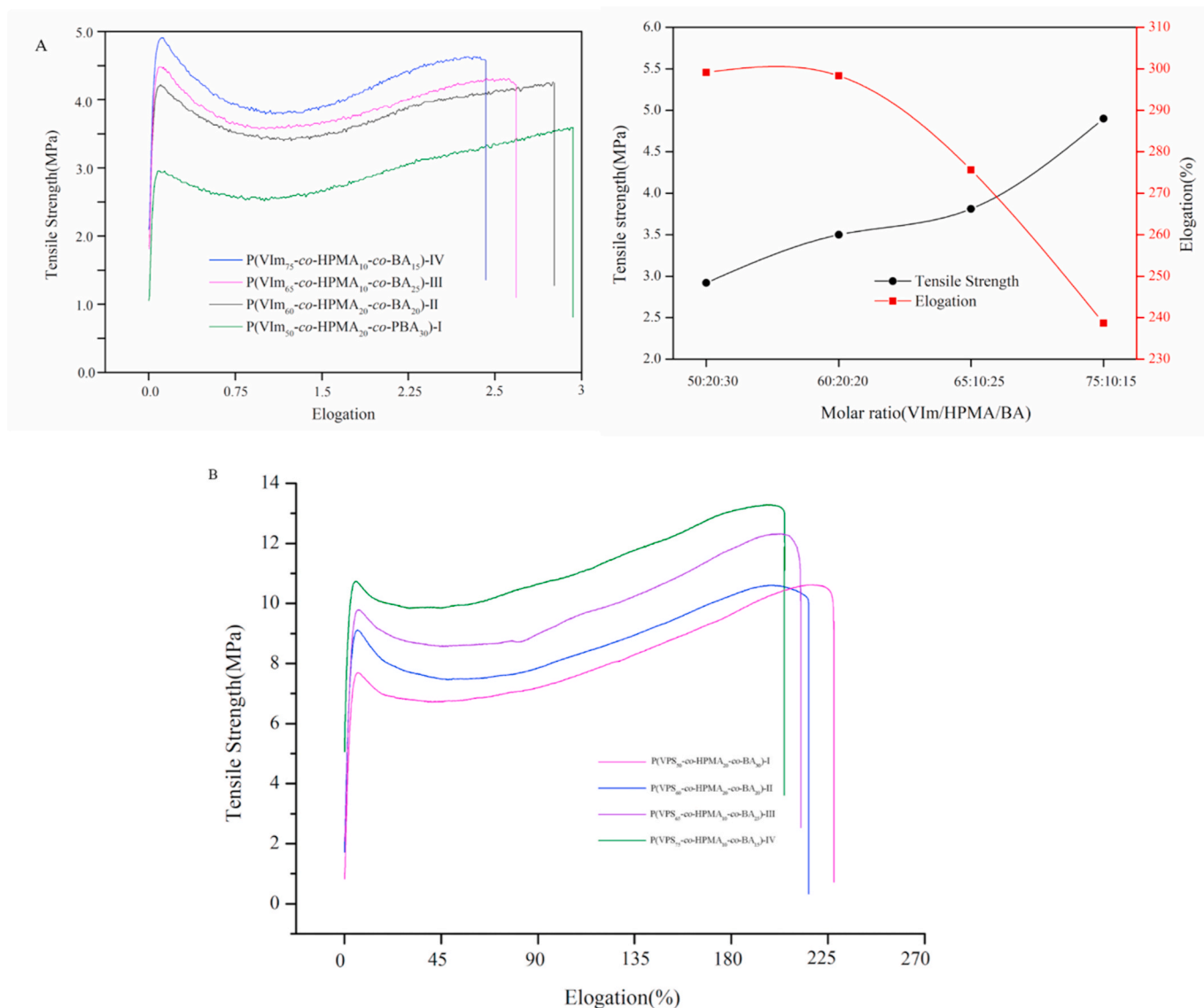


Fig. 6. The tensile strength and breaking elongation ratio of P(VIm-co-HPMA-co-BA) membranes with different molar ratios(A); The tensile strength and breaking elongation ratio of P(VPS-co-HPMA-co-BA) membranes with different molar ratios (B).

dish lid. The samples were incubated at 37 °C with 40 rpm shaking for 2 h, after 2 h, the samples were then soaked by 5 mL PBS at 37 °C with shaking gently in a sterile shaker to remove any loosely bound bacteria, the number of bacteria adhering to the membranes surface were determined under the observation of Inverted Fluorescence Microscope (LEICA DMI8, America).

2.10. Cytotoxicity test

The NIH-3T3 mouse embryonic fibroblasts were incubated for 24 h in a 48-well plate (10^4 cells/well) containing 200 μ L DMEM-high glucose (DMEM-HG) medium with 10% fetal bovine serum, the cell suspensions were then added into 48-well plate with VPS DI water solution, P(VIm-co-HPMA-co-BA) membrane, P(VPS-co-HPMA-co-BA) membranes and alpha-MEM medium (control group), which were cultured for 4 h after the culture medium was removed by PBS. Afterward, the test mediums were replaced with 20 μ L cholecystokinin octapeptide (CCK-8) solution and 200 μ L DMEM medium was re-cultured for 1.5 h at 37 °C. After 1.5 h, the optical density (OD) of each well was measured by microplate reader (ELX800 Bio Tek, America) at a wavelength of 450 nm. The cell viability of each sample was

defined by the following formula (7) [13,45–47].

$$\text{Cell viability}(\%) = \frac{OD_{\text{sample}}}{OD_{\text{control}}} \times 100\% \quad (7)$$

Where OD_{sample} and OD_{control} indicates the absorbance of the samples and control groups, severally.

3. Result and discussion

3.1. The characterization of P(VIm-co-HPMA-co-BA) and P(VPS-co-HPMA-co-BA) synthesis and characterization

The synthesis scheme of P(VIm-co-HPMA-co-BA) and P(VPS-co-HPMA-co-BA) were illustrated in Fig. 1(a) and Fig. 1(b), the detailed synthetic processes were mentioned above. The ^1H NMR spectra of P(VIm-co-HPMA-co-BA) was exhibited in Fig. 2. The FT-IR spectra of VPS, P(VIm-co-HPMA-co-BA) and P(VPS-co-HPMA-co-BA) were presented in Fig. 3. Comparing the FT-IR spectra of P(VIm-co-HPMA-co-BA) and P(VPS-co-HPMA-co-BA), the peak at 650 cm^{-1} , 1040 cm^{-1} , 1181 cm^{-1} [37] and 1650 cm^{-1} [37,48], which illustrated the appearance of amphoteric functional groups (C-S, $-\text{SO}_3^-$ and C-N^+). Detailly, the FTIR

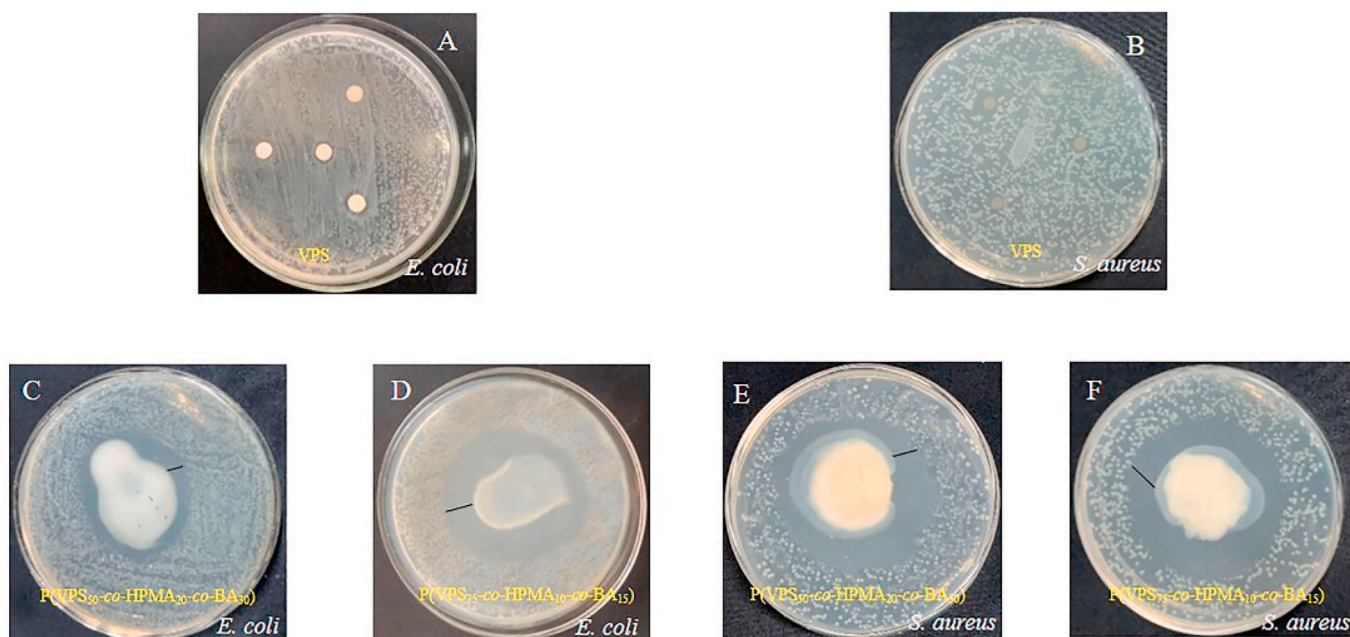


Fig. 7. The zone of inhibition tests of VPS and P(VPS-co-HPMA-co-BA) membranes (VPS (A, *E. coli*); VPS (B, *S. aureus*); P(VPS₅₀-co-HPMA₂₀-co-BA₃₀)-I (C, *E. coli*); P(VPS₅₀-co-HPMA₂₀-co-BA₂₀)-I (E, *S. aureus*); P(VPS₇₅-co-HPMA₁₀-co-BA₁₅)-IV (D, *E. coli*) and P(VPS₇₅-co-HPMA₁₀-co-BA₁₅)-IV (F, *S. aureus*).

spectrum of P(VPS-co-HPMA-co-BA) with different molar ratios of VPS/HPMA/BA indicated the intensity of the absorption peak at 1181 cm^{-1} , 1040 cm^{-1} increased. It was indicated that VPS, P(VIm-co-HPMA-co-BA) and P(VPS-co-HPMA-co-BA) were successfully prepared (see Table 1).

The molecular weights and PDI of P(VIm-co-HPMA-co-BA) with different molar ratios of VIm/HPMA/BA were shown in Table 2. The number-average molecular weight (M_n) of P(VIm-co-HPMA-co-BA) terpolymers (P(VIm₅₀-co-HPMA₂₀-co-BA₃₀)-I, P(VIm₆₀-co-HPMA₂₀-co-BA₂₀)-II, P(VIm₅₀-co-HPMA₂₀-co-BA₃₀)-V, P(VIm₆₀-co-HPMA₂₀-co-BA₂₀)-VI) were controlled in the range of 190 KDa-240KDa with the increase of HPMA monomers and BA monomers in the reaction system, furthermore, the molecular weights could be accurately controlled on the basic of feed ratios of each monomer to initiator. The elemental analysis was required to analyze the ratios of imidazole ring in P(VIm-co-HPMA-co-BA), as shown in Table 2, which could control the input of the molar ratios of 1,3-PS EtOH solution to PVIm.

3.2. Membrane surface characteristics

The degree of modification in P(VPS-co-HPMA-co-BA) with different molar ratios of VPS/HPMA/BA were not accurately represented by the FI-IR. Thus, the quaternization degree (QD) and quaternization density (QE) of samples were to analyze process of zwitterionicalization (Fig. 4), the greatest QE of 1.8 mg cm^{-2} was obtained until the reaction time was ca. 30 h in Fig. 4(a). Further notice that QD increased as reaction time and reaction temperature increased, respectively. It was an optimum reaction condition for zwitterionicalization that the reaction time reached ca. 13 h and the reaction temperature was 70 °C in Fig. 4(b). The low degree of zwitterionicalization has been attributed to steric hindrance caused by the presence of the imidazole ring [49]. It is difficult to gain the molecular weights and thicknesses of zwitterionic polymers [50], thus it is valuable to balance the reaction times and the reaction temperatures for the zwitterionicalization.

The hydrophilous membranes surface forms hydration layer on membranes surface by electrostatic action except for hydrogen bonds, which resist the adhesion of bacteria [49,50], the water contact angle of P(VIm-co-HPMA-co-BA) and P(VIm-co-HPMA-co-BA) membranes were

showed in Fig. 5. Especially when its molar ratios of zwitterionicalized PVIm (PVPS) reached 75%, the water contact angle of P(VPS-co-HPMA-co-BA) decreased to 49.6°, illustrating that the hydrophilicity ability of P(VPS-co-HPMA-co-BA) membranes surface was improved by increasing the molar ratios of PVPS in P(VPS-co-HPMA-co-BA).

The tensile strength and breaking elongation ratio of P(VIm-co-HPMA-co-BA) membranes with different molar ratios of VIm/HPMA/BA were tested (Fig. 6(A)). The tensile strength of the P(VIm-co-HPMA-co-BA) membranes gradually increased along with the increase of the PVIm content in P(VIm-co-HPMA-co-BA), and the elongation at break of P(VIm-co-HPMA-co-BA) membranes gradually decreased, detailly, the results suggested that tensile strength and breaking elongation could reach 3.81 MPa, 275.6%, severally, when molar ratios of VIm/HPMA/BA in P(VIm-co-HPMA-co-BA) were controlled in 65/10/25. The introduction of PVPS sharply enhanced the tensile strength of P(VPS-co-HPMA-co-BA) membranes, the highest tensile strength could reach 10.89 MPa, which were shown in Fig. 6(B). A sufficient reason was that soft PBA network could be used as a polymer network [28,32] and the imidazole ring in P(VIm-co-HPMA-co-BA) reduced its stiffness [14,51], which made a great effect on the mechanical properties of above samples.

3.3. Antimicrobial activity and antifouling assessment of sample membranes

3.3.1. Antibacterial activity of sample membranes

The anti-bacterial activities of VPS DI solution and P(VPS₅₀-co-HPMA₂₀-co-BA₃₀) and P(VPS₇₅-co-HPMA₁₀-co-BA₁₅) membranes were evaluated according to previous study [52,53]. The antibacterial activity of VPS and P(VPS-co-HPMA-co-BA) membranes that cultivated with *E. coli* and *S. aureus* was measured by zone of inhibition tests (Fig. 7), the diameters of inhibition zone of VPS were 9 mm for *S. aureus* and 6 mm for *E. coli*, respectively, which showed the weak antibacterial activity of VPS against *E. coli*. Simultaneously, P(VPS₅₀-co-HPMA₂₀-co-BA₃₀) membrane was cultivated with *E. coli* and *S. aureus* with 17.2 ± 0.50 mm, 22.6 ± 0.45 mm diameter, respectively, the diameter of inhibition zone of P(VPS₇₅-co-HPMA₁₀-co-BA₁₅) membrane against *E. coli* and *S. aureus* were 19.3 ± 0.40 mm, 25.8 ± 0.60 mm, respectively.

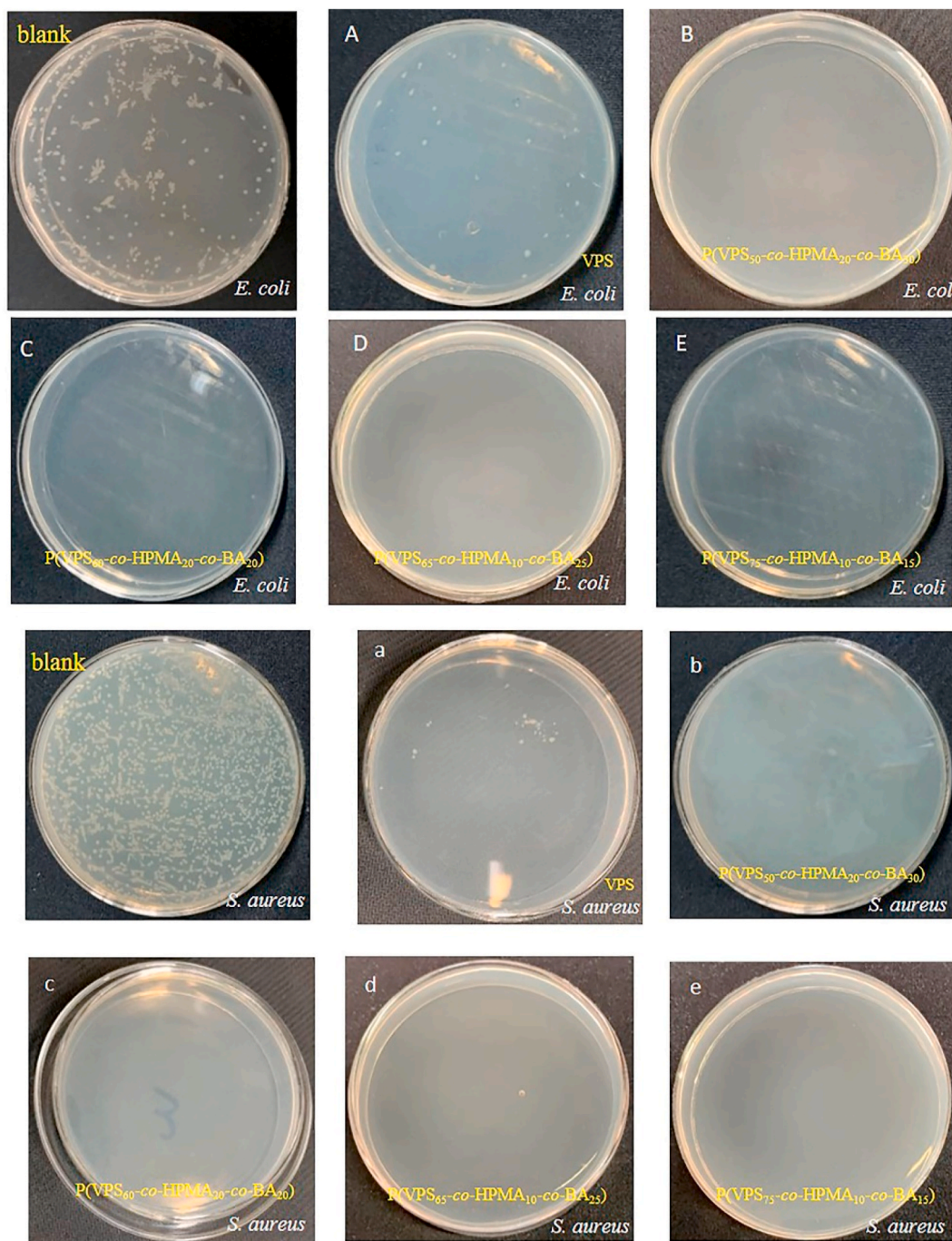


Fig. 8. The images of bacterial colonies of VPS DI water solution (20 mg/mL) (A, a), P(VPS₅₀-co-HPMA₂₀-co-BA₃₀)-I (B, b), P(VPS₆₀-co-HPMA₂₀-co-BA₂₀)-II (C, c), P(VPS₆₅-co-HPMA₁₀-co-BA₂₅)-III (D, d), P(VPS₇₅-co-HPMA₁₀-co-BA₁₅)-IV (E, e) membranes (3.5 cm²).

The images of bacterial colonies of *E. coli* (A-E) and *S. aureus* (a-e) were presented on solid medium that was cultivated with VPS DI water solution (20 mg/mL) (A, a), the concentration of the solution was based on the corresponding literature [54,55], P(VPS₅₀-co-HPMA₂₀-co-BA₃₀)-I (B, b), P(VPS₆₀-co-HPMA₂₀-co-BA₂₀)-II (C, c), P(VPS₆₅-co-HPMA₁₀-co-BA₂₅)-III (D, d), P(VPS₇₅-co-HPMA₁₀-co-BA₁₅)-IV (E, e) membranes (3.5 cm²) were shown in Fig. 8. Detailly, much smaller number of *E. coli* and *S. aureus* colonies grew on the solid mediums when P(VPS-co-HPMA-co-BA) membranes were cultivated with them than VPS, further, the relative bactericidal rates of P(VPS-co-HPMA-co-BA) membranes and VPS cultivated with *E. coli* and *S. aureus* were counted in Fig. 9, the livability of bacterium of PVPS-co-HPMA-co-PBA membranes against *E. coli* and *S. aureus* were visibly much higher than VPS with 99.99%,

averagely.

3.3.2. Antibacterial mechanism

The P(VPS-co-HPMA-co-BA) membranes, its designed PVPS content was 50% and 75%, were chosen to experimental samples. There was an increase of the absorbance changes at 420 nm within 70 min, 50 min, respectively in Fig. 10, the result of which mainly attributed to the increasing of designed content of PBA in a three-component polymer system, evaluating that a unique hydrolysis of the galactosyl bond in ONPG occurs during the reaction of the beta-galactosidase of *E. coli*. The results indicated that P(VPS-co-HPMA-co-BA) membranes could destroy the integrity of *E. coli* membranes.

The morphological changes of *E. coli* and *S. aureus* were

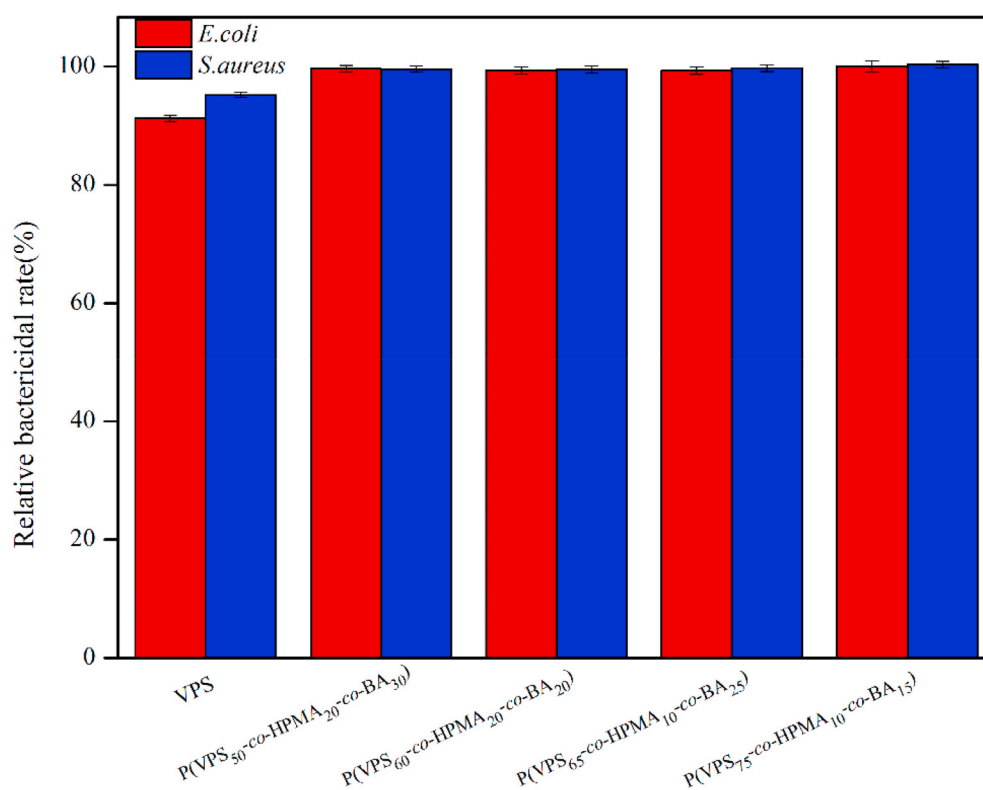


Fig. 9. The relative bactericidal rates of PVPS-co-HPMA-co-PBA membranes (P(VPS₅₀-co-HPMA₂₀-co-BA₃₀)-I; P(VPS₆₀-co-HPMA₂₀-co-BA₂₀)-II; P(VPS₆₅-co-HPMA₁₀-co-BA₂₅)-III and P(VPS₇₅-co-HPMA₁₀-co-BA₁₅)-IV) and VPS DI Solution (20 mg/mL).

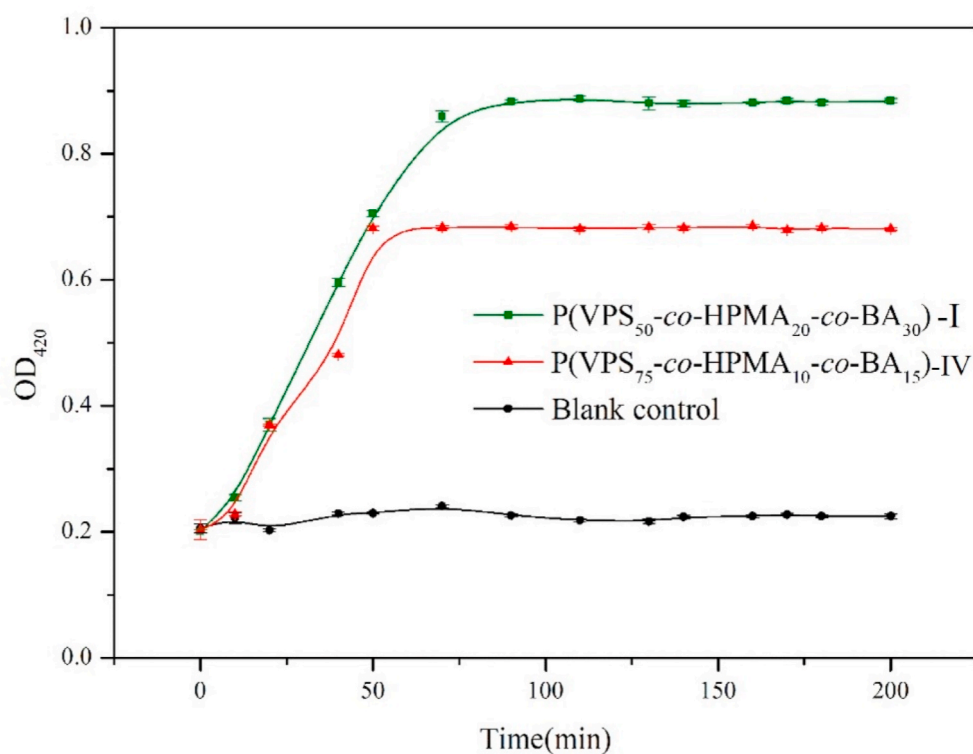


Fig. 10. The absorbance changes of *o*-nitrophenol (ONP) incubated with blank control, P(VPS₅₀-co-HPMA₂₀-co-BA₃₀)-I and P(VPS₇₅-co-HPMA₁₀-co-BA₁₅)-IV membranes as times changed at 420 nm.

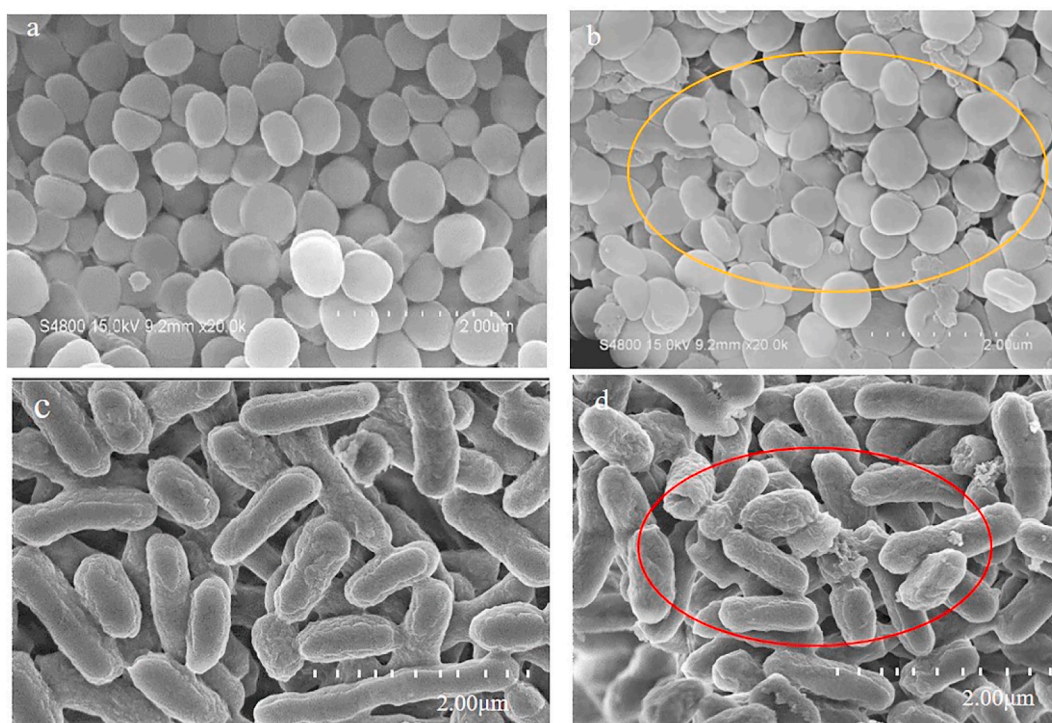


Fig. 11. The morphological changes of *E. coli* and *S. aureus* (control samples (a, c), P(VPS-co-HPMA-co-BA) membranes (b, d)).

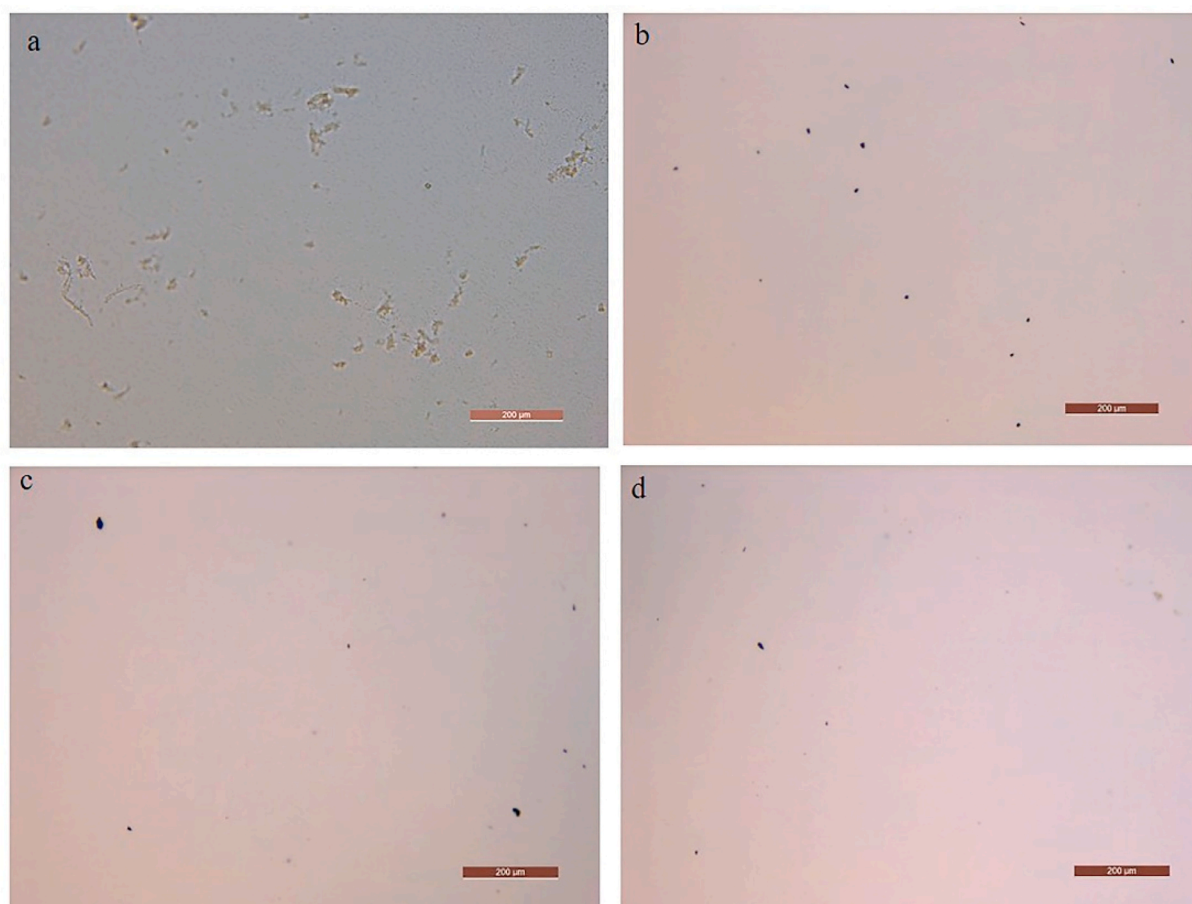


Fig. 12. Bacterial surface adhesion on P(VPS-co-HPMA-co-BA) membranes and polystyrene surfaces (PS control group (a), P(VPS₇₅-co-PHPMA₁₀-co-PBA₁₅)-IV (b), P(VPS₆₀-co-PHPMA₂₀-co-PBA₂₀)-II (c), P(VPS₅₀-co-PHPMA₂₀-co-PBA₃₀)-I (d)).

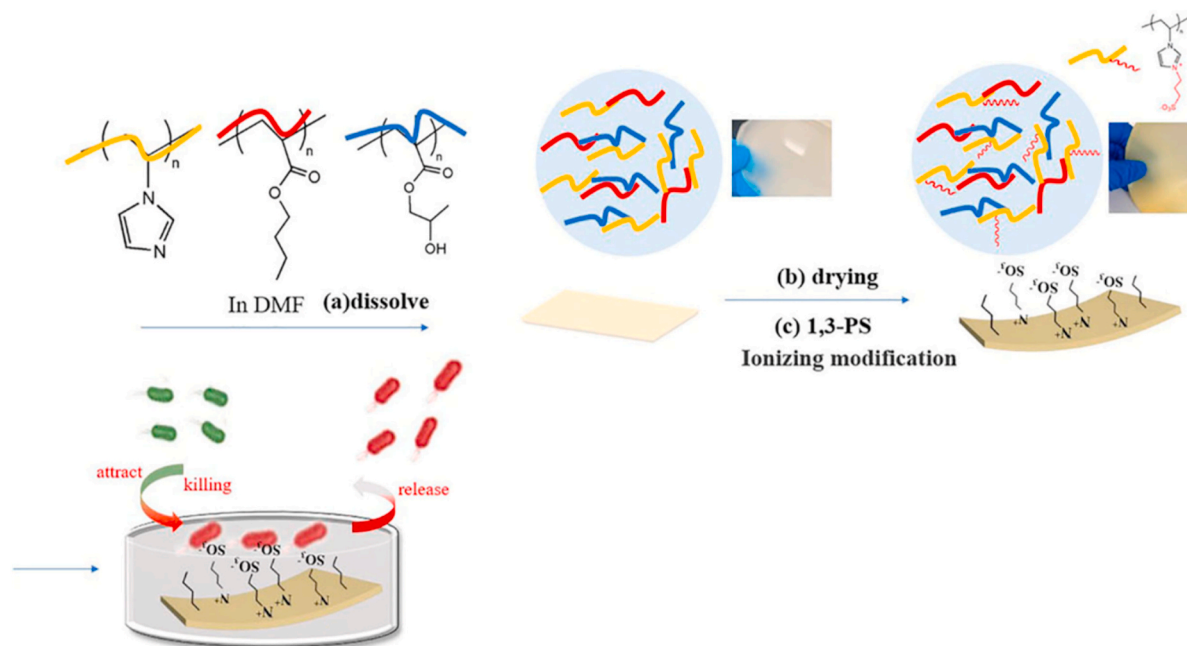


Fig. 13. The diagram of antibacterial and antifouling activity of P(VPS-co-HPMA-co-BA) membranes.

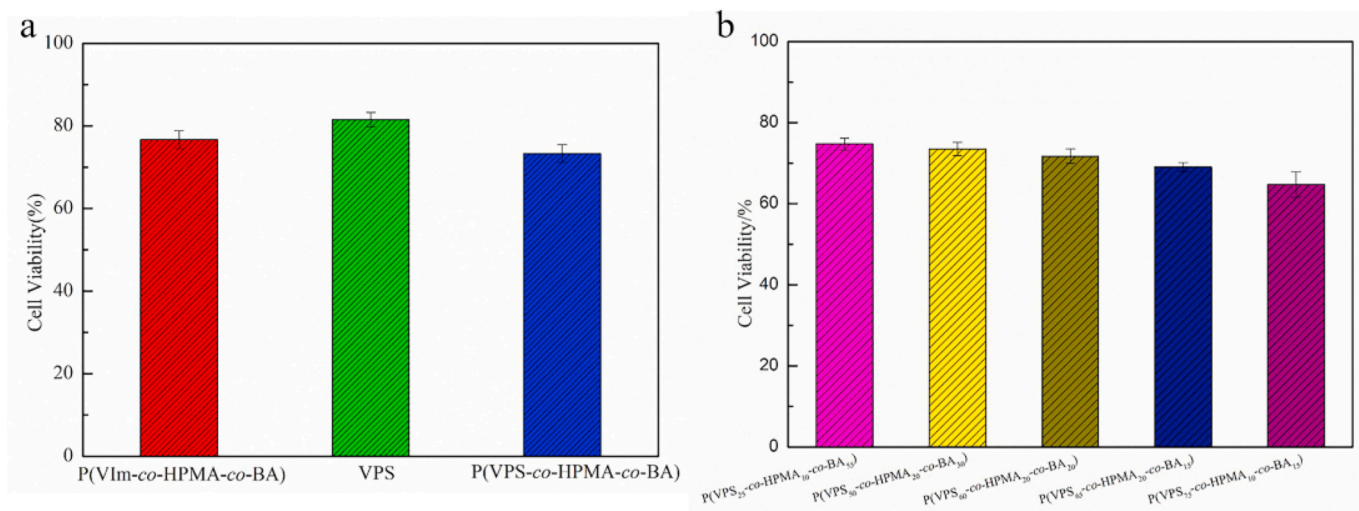


Fig. 14. The cell survival rates of VPS; P(VIm₆₅-co-HPMA₁₀-co-BA₂₅)-III, P(VPS₆₅-co-HPMA₁₀-co-BA₂₅)-VII membranes (a) and P(VPS-co-HPMA-co-BA) membranes with different molar ratios of VPS/HPMA/BA (b).

characterized by SEM (Fig. 11). The *S. aureus* and *E. coli* from the control samples displayed smooth and flat, rod-shaped structure, respectively (Fig. 11(a) and (c)), however, the bacterial cellular membranes showed obvious abnormal and defective morphology (Fig. 11(b) and (d)) after incubated with P(VPS-co-HPMA-co-BA) membranes. The above results indicated that the antibacterial activity of P(VPS-co-HPMA-co-BA) membranes were realized by puncturing *E. coli* and *S. aureus* membranes to release intracellular ingredients. The P(VPS-co-HPMA-co-BA) contains positive ions and alkyl chains groups just like a “thorn” in a specific system, which could puncture bacterial membrane, destroy the normal cell morphology and perturb regular bacterial physiological activities [31].

3.3.3. Bacterial surface adhesion assay

According to corresponding study [10–12] and the authors made a slight redesign based on the previous experimental standards. Bacterial

surface adhesion on P(VPS-co-HPMA-co-BA) membranes and polystyrene surfaces were shown on Fig. 12. Compared with the polystyrene surface for control, P(VPS-co-HPMA-co-BA) membranes had obvious anti-adhesion property based on the display for bacteria population density, it was further found that the smaller numbers of bacteria adhered to the membranes surface with the increase of molar ratio of PVPS groups in P(VPS-co-HPMA-co-BA), a reasonable explanation was that PVPS groups easily formed energy barrier and hydration layer attached on the membranes surface was susceptible to be formed, which effectively prevented bacteria from adhering to its surface.

The diagram for antibacterial and antifouling activity of P(VPS-co-HPMA-co-BA) membranes was shown in Fig. 13.

3.4. Cytotoxicity

For the above-mentioned samples exhibited antibacterial property,

the cytotoxicity of samples has a great influence on its application for environmental protection. The cell survival rates of NIH-3T3 cells cultured with VPS DI solution (20 mg/mL); P(VIm₆₅-co-HPMA₁₀-co-BA₂₅)-III and P(VPS₆₅-co-HPMA₁₀-co-BA₂₅)-VII membranes were showed in Fig. 14, the cell viability of three samples reached 70–80% (Fig. 14(a)), As shown in Fig. 14(b), with the increasing of PBA content equipped with biocompatible property in P(VPS-co-HPMA-co-BA), the cell survive rate came up to 73%–78% that exhibited great non-toxicity [13,47], nevertheless, the cell survive rate of P(VPS₇₅-co-HPMA₁₀-co-BA₁₅) membrane reduced to 62% because of the cytotoxicity of N⁺ group in P(VPS-co-HPMA-co-BA). Overall, the molar ratios of each segment of P(VIm-co-HPMA-co-BA) should be controlled before and after zwitterionization, which could measure non-toxicity of P(VPS-co-HPMA-co-BA). The above results showed P(VPS-co-HPMA-co-BA) membranes were non-toxicity and had a good prospect for more applications.

4. Conclusions

In this article we have prepared the novel P(VIm-co-HPMA-co-BA) and P(VPS-co-HPMA-co-BA). The optimal DQ of 1.8 mg cm⁻² was gained until the reaction time reached 30 h, specifically, it was the reaction time reached ca.13 h and the reaction temperature was 70 °C that were optimal conditions for the zwitterionization. Meanwhile, the antibacterial property assessments and bacterial adhesion assay results showed that the inhibition zone diameters of P(VPS-co-HPMA-co-BA) membranes were bigger than VPS, and the sterilization rates of P(VPS-co-HPMA-co-BA) membranes against *E. coli* and *S. aureus* were 99.99% averagely. Furthermore, it was much better *anti*-adhesion property with the increase of the molar ratios of PVPS in P(VPS-co-HPMA-co-BA). Then the improved mechanical properties of P(VPS-co-HPMA-co-BA) membranes have been promoted by ionic bonding of PVPS groups, specifically, the maximum tensile strength could reach 10.9 MPa. Finally, the cytotoxicity experiments had shown the non-toxicity of P(VPS-co-HPMA-co-BA) membranes, the cell survival rates were between 70 and 80%. The above results exhibited that P(VPS-co-HPMA-co-BA) had a potential application in marine coatings.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] D. Davies, Understanding biofilm resistance to antibacterial agents, *Nat. Rev. Drug Discov.* 2 (2) (2003) 114–122.
- [2] B.Q. Liao, D.M. B. H.E. Kraemer, G.G. Leppard, S.N. Liss, A review of biofouling and its control in membrane separation bioreactors, *Water Environ. Res.* 76 (5) (2004).
- [3] A.J. De Kerchove, M. Elimelech, Impact of alginate conditioning film on deposition kinetics of motile and nonmotile *Pseudomonas aeruginosa* strains, *Appl. Environ. Microbiol.* 73 (16) (2007) 5227–5234.
- [4] M. Ben-Sasson, K.R. Zdrov, Q. Geng, et al., Surface functionalization of thin-film composite membranes with copper nanoparticles for antimicrobial surface properties, *Environ. Sci. Technol.* 48 (1) (2014) 384–393.
- [5] S.E. Kwan, E. Bar-Zeev, M. Elimelech, Biofouling in forward osmosis and reverse osmosis: measurements and mechanisms, *J. Membr. Sci.* 493 (2015) 703–708.
- [6] B. Andrew, M.V. Lowe, Michael A. Wassall, Lavinia Wong, Norman C. Billingham, P. Steven, A.W.L. Armes, Well-defined sulfobetaine-based statistical copolymers as potential antibioadherent coatings, *Poly(Sulfobetaine) Copolymers As Potential Antibioadherent Coat.* (2000) 89–93.
- [7] J. Landoulsi, K.E. Cooksey, V. Dupres, Review–Interactions between diatoms and stainless steel: focus on biofouling and biocorrosion, *Biofouling* 27 (10) (2011) 1109–1124.
- [8] C. Diamond, The market for marine and protective coatings is being driven by an increase in ship repair and maintenance activities globally, *Coating World* (2016).
- [9] Q. Xie, J. Pan, C. Ma, et al., Dynamic surface antifouling: mechanism and systems, *Soft Matter* 15 (6) (2019) 1087–1107.
- [10] G. Cheng, G. Li, H. Xue, et al., Zwitterionic carboxybetaine polymer surfaces and their resistance to long-term biofilm formation, *Biomaterials* 30 (28) (2009) 5234–5240.
- [11] L. Mi, S. Jiang, Synchronizing nonfouling and antimicrobial properties in a zwitterionic hydrogel, *Biomaterials* 33 (35) (2012) 8928–8933.
- [12] H. Salehi, M. Rastgar, A. Shakeri, Anti-fouling and high water permeable forward osmosis membrane fabricated via layer by layer assembly of chitosan/graphene oxide, *Appl. Surf. Sci.* 413 (2017) 99–108.
- [13] L. Huang, L. Zhang, S. Xiao, et al., Bacteria killing and release of salt-responsive, regenerative, double-layered polyzwitterionic brushes, *Chem. Eng. J.* 333 (2018) 1–10.
- [14] H. Caner, E. Yilmaz, O. Yilmaz, Synthesis, characterization and antibacterial activity of poly(N-vinylimidazole) grafted chitosan, *Carbohydr. Polym.* 69 (2) (2007) 318–325.
- [15] M.W. Sabaa, N.A. Mohamed, R.R. Mohamed, et al., Synthesis, characterization and antimicrobial activity of poly (N-vinyl imidazole) grafted carboxymethyl chitosan, *Carbohydr. Polym.* 79 (4) (2010) 998–1005.
- [16] A.T. Nguyen, J. Baggerman, J.M. Paulusse, et al., Stable protein-repellent zwitterionic polymer brushes grafted from silicon nitride, *Langmuir* 27 (6) (2011) 2587–2594.
- [17] J.B. Schlenoff, Zwitterion: coating surfaces with zwitterionic functionality to reduce nonspecific adsorption, *Langmuir* 30 (32) (2014) 9625–9636.
- [18] G. Ye, J. Lee, F. Perreault, et al., Controlled architecture of dual-functional block copolymer brushes on thin-film composite membranes for integrated "defending" and "attacking" strategies against biofouling, *ACS Appl. Mater. Interfaces* 7 (41) (2015) 23069–23079.
- [19] I. Banerjee, R.C. Pangule, R.S. Kane, Antifouling coatings: recent developments in the design of surfaces that prevent fouling by proteins, bacteria, and marine organisms, *Adv. Mater.* 23 (6) (2011) 690–718.
- [20] A.M. Carmona-Ribeiro, L.D. De Melo Carrasco, Cationic antimicrobial polymers and their assemblies, *Int. J. Mol. Sci.* 14 (5) (2013) 9906–9946.
- [21] M.L. Marré Tirado, M. Bass, M. Piatkovsky, et al., Assessing biofouling resistance of a polyamide reverse osmosis membrane surface-modified with a zwitterionic polymer, *J. Membr. Sci.* 520 (2016) 490–498.
- [22] S. Colak, G.N. Tew, Dual-functional ROMP-based betaines: effect of hydrophilicity and backbone structure on nonfouling properties, *Langmuir* 28 (1) (2012) 666–675.
- [23] Y. Shi, Z. Fu, B. Li, et al., Synthesis of diblock copolymers by combining stable free radical polymerization and atom transfer radical polymerization, *J. Polym. Sci. Polym. Chem.* 44 (8) (2006) 2468–2475.
- [24] Veerle M.C. Coessens, Krzysztof Matyjaszewski, Fundamentals of atom transfer radical polymerization, *J. Chem. Educ.* 87 (9) (2010) 916–919.
- [25] A.K. A. Matyjaszewski*, Simultaneous EPR and kinetic study of styrene atom transfer radical polymerization (ATRP), *Macromolecules* 31 (1998) 5695–5701.
- [26] W.-W. Yue, H.-J. Li, T. Xiang, et al., Grafting of zwitterion from polysulfone membrane via surface-initiated ATRP with enhanced antifouling property and biocompatibility, *J. Membr. Sci.* 446 (2013) 79–91.
- [27] L. Mazzocchi, M. Scandola, Z. Jiang, Random copolymerization with a large lactone enhances aliphatic polycarbonate crystallinity, *Eur. Polym. J.* 48 (11) (2012) 1883–1891.
- [28] J. Cui, K. Kratz, B. Hiebl, et al., Soft poly(n-butyl acrylate) networks with tailored mechanical properties designed as substrates for in vitro models, *Polym. Adv. Technol.* 22 (1) (2011) 126–132.
- [29] Andreas Lendlein, A.M. S. R. L. AB-polymer networks based on oligo (ε-caprolactone) segments showing shape-memory properties, *Natl. Acad. Sci.* 98 (2001) 842–847.
- [30] M. Gauthier, T. Carrozzella, A. Penlidis, Sulfobetaine zwitterionomers based on n-butyl acrylate and 2-ethoxyethyl acrylate: monomer synthesis and copolymerization behavior, *J. Polym. Sci. Polym. Chem.* 40 (4) (2002) 511–523.
- [31] J. Jung, Y. Bae, Y. Kwan Cho, et al., Structural insights into conformation of amphiphilic quaternary ammonium chitosans to control fungicidal and anti-biofilm functions, *Carbohydr. Polym.* 228 (2020), 115391.
- [32] B. Hiebl, J. Cui, K. Kratz, et al., Viability, morphology and function of primary endothelial cells on poly(n-butyl acrylate) networks having elastic moduli comparable to arteries, *J. Biomater. Sci. Polym. Ed.* 23 (7) (2012) 901–915.
- [33] X. Zhang, F. Shi, J. Niu, et al., Superhydrophobic surfaces: from structural control to functional application, *J. Mater. Chem.* 18 (6) (2008) 621–633.
- [34] G. Alexander, R.B.K. Didukh, Larisa A. Bimendina, Sarkyt E. Kudaibergenov, Synthesis and characterization of novel hydrophobically modified polybetaines as pour point depressants, *Novel Hydrophobically Modified Polybetaines* 92 (2004) 1042–1048.
- [35] H. Zou, S.P. Armes, Efficient synthesis of poly(2-hydroxypropyl methacrylate)-silica colloidal nanocomposite particles via aqueous dispersion polymerization, *Polym. Chem.* 3 (1) (2012) 172–181.

- [36] Y.-F. Zhao, L.-P. Zhu, Z. Yi, et al., Improving the hydrophilicity and fouling-resistance of polysulfone ultrafiltration membranes via surface zwitterionization mediated by polysulfone-based triblock copolymer additive, *J. Membr. Sci.* 440 (2013) 40–47.
- [37] X. Ma, Z. Zhu, W. Shi, et al., Synthesis and application of a novel betaine-type copolymer as fluid loss additive for water-based drilling fluid, *Colloid Polym. Sci.* 295 (1) (2016) 53–66.
- [38] Kazunori Se, Masato Kijima, Reiko Ohtomo, Teruo Fujimoto, Quaternization of poly(tertiary aminostyrene)s and characterization of the quaternized polymers, *Quat. Poly(Tertiary Aminostyrene)s* (1996) 1219–1226.
- [39] F. Chen, Z. Sun, Preparation of homogeneous grafting cellulose and partial substitution for polyethersulfone membrane material, *Carbohydr. Polym.* 95 (1) (2013) 85–90.
- [40] N.B. Hayal Bulbul Sonmez, Quaternization of poly(4-vinyl pyridine) beads with 2-chloroacetamide for selective mercury extraction, *React. Funct. Polym.* 51 (2002) 55–60.
- [41] S. Zhang, C. Xia, Y. Dong, et al., Soy protein isolate-based films reinforced by surface modified cellulose nanocrystal, *Ind. Crop. Prod.* 80 (2016) 207–213.
- [42] P. Roach, N.J. Shirtcliffe, M.I. Newton, Progress in superhydrophobic surface development, *Soft Matter* 4 (2) (2008) 224–240.
- [43] S.C. Zheng Zhang, Yung Chang, Shaoyi Jiang, Surface grafted sulfobetaine polymers via atom transfer radical polymerization as superlow fouling coatings, *J. Phys. Chem. B* 110 (2006) 10799–10804.
- [44] P. Fei, L. Liao, J. Meng, et al., Synthesis, characterization and antibacterial properties of reverse osmosis membranes from cellulose bromoacetate, *Cellulose* 25 (10) (2018) 5967–5984.
- [45] I. Singh, A. Priyam, D. Jha, et al., Polydopamine -aminoglycoside nanoconjugates: synthesis, characterization, antimicrobial evaluation and cytocompatibility, *Mater Sci Eng C Mater Biol Appl* 107 (2020), 110284.
- [46] M. Divya, B. Vaseeharan, M. Abinaya, et al., Biopolymer gelatin-coated zinc oxide nanoparticles showed high antibacterial, antibiofilm and anti-angiogenic activity, *J. Photochem. Photobiol., B* 178 (2018) 211–218.
- [47] S.M. Caroline Diana, M.R. Rekha, Efficacy of vinyl imidazole grafted cationized pullulan and dextran as gene delivery vectors: a comparative study, *Int. J. Biol. Macromol.* 105 (Pt 1) (2017) 947–955.
- [48] C. Panawong, S. Martwiset, Synthesis and characterization of poly(styrene sulfonic acid-co-1-vinylimidazole-co-styrene) and its blends with poly(vinyl chloride) as proton conducting membranes, *Polym. Bull.* 75 (9) (2017) 3843–3858.
- [49] S. Turmanova, M. Minchev, K. Vassilev, et al., Surface grafting polymerization of vinyl monomers on poly(tetrafluoroethylene) films by plasma treatment, *J. Polym. Res.* 15 (4) (2008) 309–318.
- [50] F.J. Xu, S.P. Zhong, L.Y. Yung, et al., Thermoresponsive comb-shaped copolymer-Si (100) hybrids for accelerated temperature-dependent cell detachment, *Biomaterials* 27 (8) (2006) 1236–1245.
- [51] G. Garg, G.S. Chauhan, R. Gupta, et al., Anion effects on anti-microbial activity of poly[1-vinyl-3-(2-sulfoethyl imidazolium betaine)], *J. Colloid Interface Sci.* 344 (1) (2010) 90–96.
- [52] X. Chen, G. Zhang, Q. Zhang, et al., Preparation and performance of amphiphilic polyurethane copolymers with capsaicin-mimic and PEG moieties for protein resistance and antibacteria, *Ind. Eng. Chem. Res.* 54 (15) (2015) 3813–3820.
- [53] Y. Chen, Y. Ding, J. Zheng, A polymer nanocomposite coating with enhanced hydrophilicity, antibacterial and antibiofouling properties: role of polymerizable emulsifier/anionic ligand, *Chem. Eng. J.* 379 (2020), 122268.
- [54] Y. Xie, L. Chen, X. Zhang, et al., Integrating zwitterionic polymer and Ag nanoparticles on polymeric membrane surface to prepare antifouling and bactericidal surface via Schiff-based layer-by-layer assembly, *J. Colloid Interface Sci.* 510 (2018) 308–317.
- [55] S. Zhang, C. Kai, B. Liu, et al., Facile fabrication of cellulose membrane containing polyiodides and its antibacterial properties, *Appl. Surf. Sci.* 500 (2020), 144046.